DRAFT — P1–P5 MEASURED; P4-a-block-energy HIT

Assignment 3: Results, Intuition & Quantitative Properties

CS 312 · Optimization 2 · **Hand-in PDF #1** (of 2).
Companion practice quiz → `a3-practice.md` / `media/pdfs/a3-practice.pdf`.

Label discipline: **MEASURED** = numbers from `media/p1/*`, `media/p2/*`, `media/p4/*`, `media/p5/*`; **FITTED** = curves on that grid (never "measured"); **PREDICTION** = locked pre-run bands; **HIT/MISS** only after MEASURED. Every float below is copied file-exact from store evidence — no invention.

Consolidated from: assignment report · experimental appendix highlights · predictive intuition · quantitative prediction reference. Older five-PDF split is **superseded** (see `a3-deliverables-readme.md`).

Setup

- d8 transformer, next-token prediction; val loss ↓ better; AdamW unless overridden (P3 full-batch GD; P4(c) SGDM).
- ~614M tokens unless a helper scales down.
- Starter: `experiments/a3_optimization/` (`a3_release` @ `57c60bb`); Modal `cs312-daxton`; W&B `daxgute-stanford-university` / `assignments`.
- Batch 1 (P1): **DONE** 30/30 cells in `media/p1/a3-mode-connectivity.json`.
- Batch 2 (P2+P3): **9/9 DONE** (`media/p2/batch2_finals.csv`).
- Batch 3 (P4): stage-a **13/13** + b/c scored; P5 cooldown **MEASURED**.

Problem 1: Mode Connectivity

Config: `wsd0.0`, peak LR `0.003`, (x\in{0,50,100,250,500,1000}), (y\in{25,50,100,250,500}), `batch_order_seed∈{1,2}).

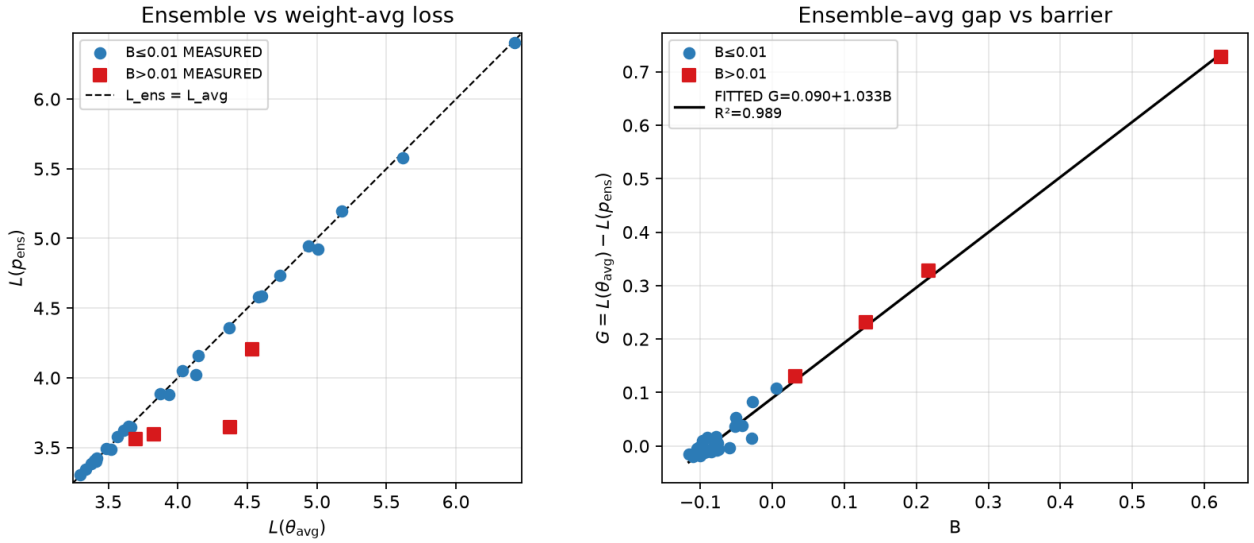
Mechanism / intuition. Under constant post-warmup LR, late training keeps large bounce amplitude (river-valley; P2). Forking copies weights + Adam state at step (x); different batch orders accumulate wall-orbit disagreement even when endpoint losses match. Longer shared prefix synchronizes trajectories before the fork, so midpoint weight-average stays in-basin; short (x) + long (y) is where the midpoint ridge appears. Ensemble averaging is in **function space** — two nets can disagree in (\theta) yet agree on soft predictions, so (L(p_{\mathrm{ens}})) can stay low while (L(\theta_{\mathrm{avg}})) sits on a ridge.

Prefix (MEASURED): `model-d8-lr0.003-tok614M-wsd0.0-stop1000` / W&B `rq1k0bz1` — end train `3.43004`, val `3.48832` (1000 steps).
Grid source: `media/p1/barrier-table.csv`, `barrier-summary.json`, `a3-mode-connectivity.json`.

(a) Endpoints + ensemble

Cell	(L_1)	(L_2)	(L_{\mathrm{ens}})	(L(\theta_{\mathrm{avg}}))
((0,500))	3.7511	3.7483	3.6454	4.3734
((1000,25))	3.4786	3.4796	3.4221	3.4194

Where (B>0.01), ensemble sits well below weight-average — parameter midpoint is bad while averaged *predictions* remain good. MEASURED failure gaps (G=L(\theta_{\mathrm{avg}})-L(p_{\mathrm{ens}})): ((100,500),0.1311); ((50,500),0.2309); ((0,250),0.3273); ((0,500),0.7281); median (G\approx 0.279).

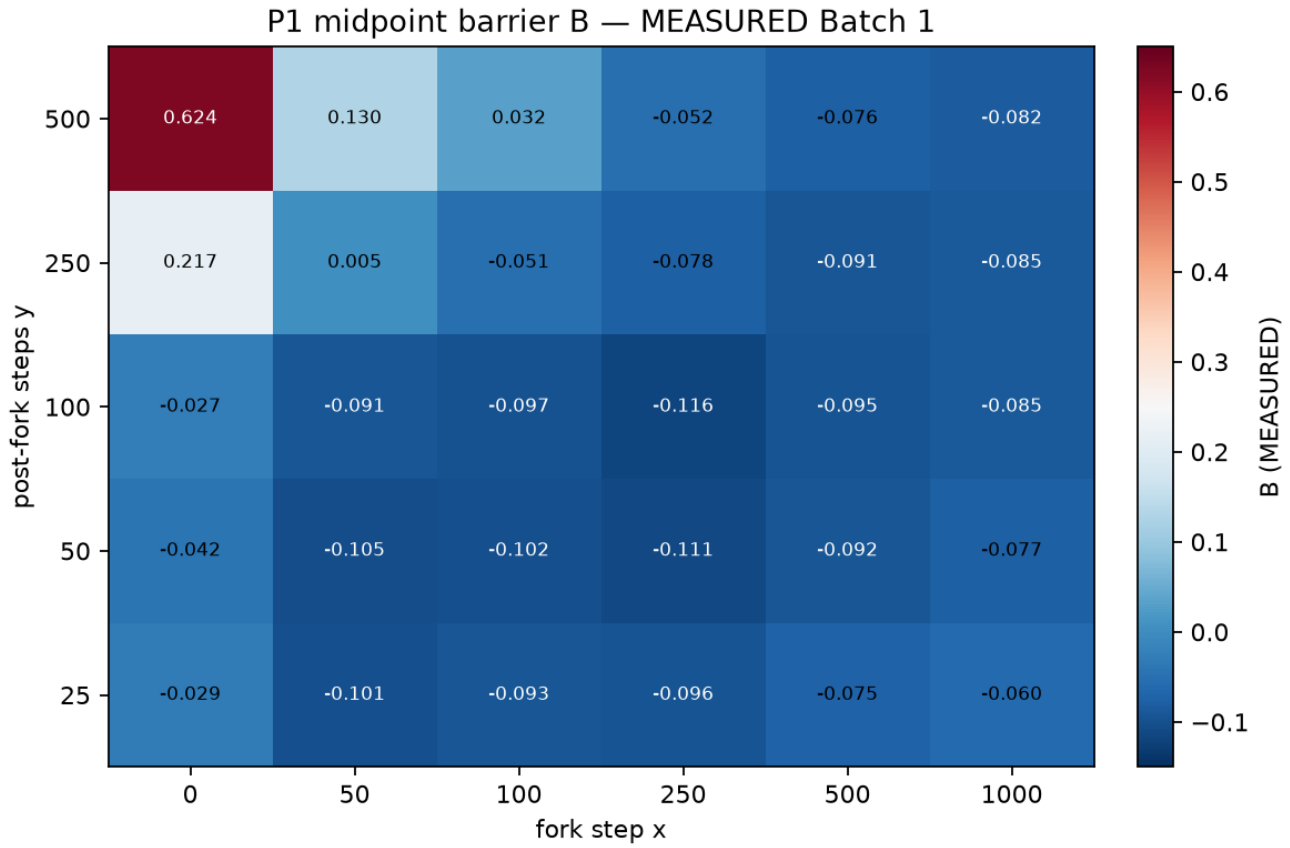


(b) Barrier (B)

[B=L(\theta_{\mathrm{avg}})-\tfrac{L(\theta_1)+L(\theta_2)}{2},\qquad \theta_{\mathrm{avg}} =\tfrac12(\theta_1+\theta_2)] at (\alpha=0.5). Handout: averaging "works" if (B\le 0.01).

MEASURED (`barrier-summary.json`, file-exact): (B_{\min}=-0.1160578727722168) @ ((250,100)); (B_{\max}=0.6237311363220215) @ ((0,500)); mean (B=-0.033404584725697836); **26/30** cells with (B\le 0.01).
Failures ((B>0.01)): ((0,250)=0.2173), ((0,500)=0.6237), ((50,500)=0.1299), ((100,500)=0.0319). Every (x\ge 250) cell has (B\le 0.01).

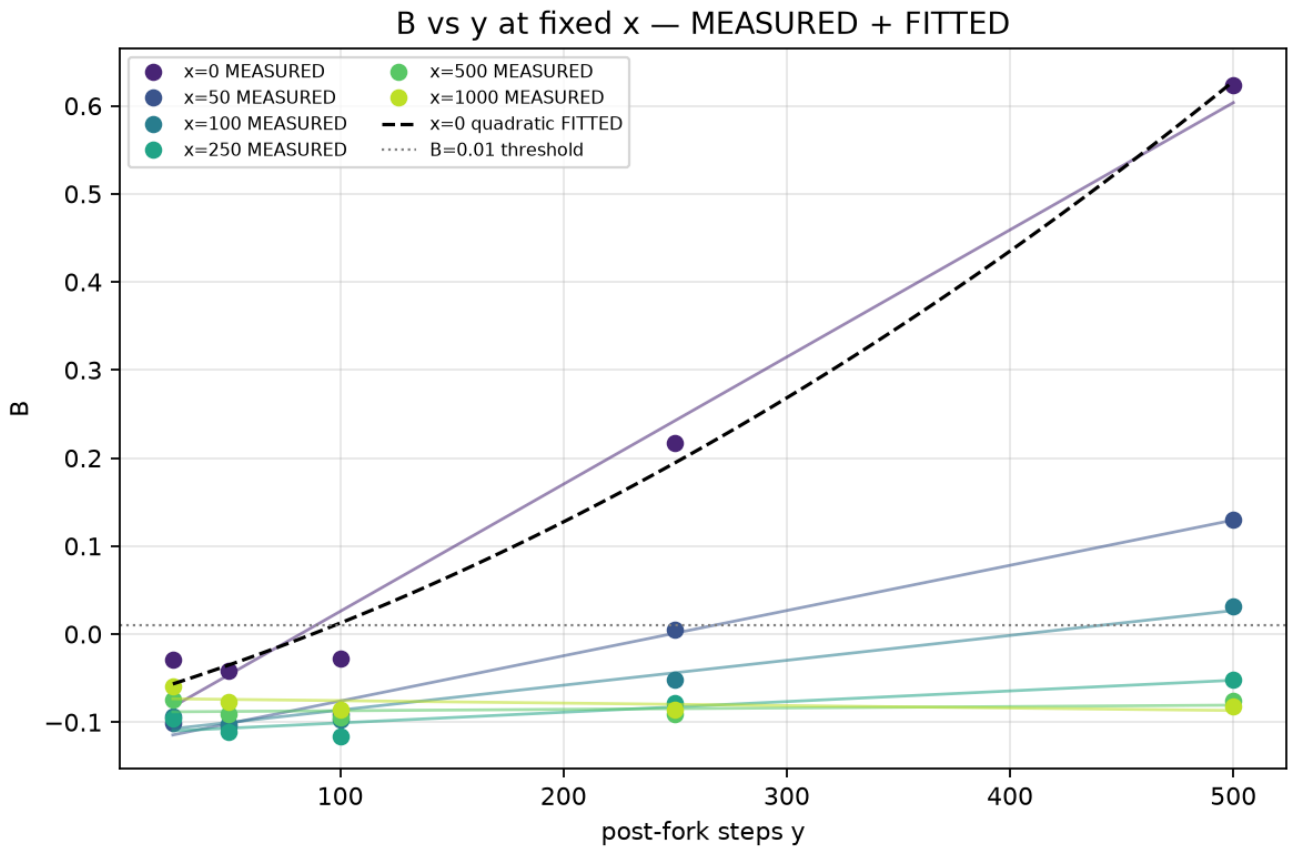
(x) \ (y)	25	50	100	250	500
0	−0.0288 ✓	−0.0416 ✓	−0.0275 ✓	0.2173	0.6237
50	−0.1009 ✓	−0.1053 ✓	−0.0910 ✓	0.0055 ✓	0.1299
100	−0.0932 ✓	−0.1019 ✓	−0.0972 ✓	−0.0512 ✓	0.0319
250	−0.0963 ✓	−0.1106 ✓	−0.1161 ✓	−0.0779 ✓	−0.0522 ✓
500	−0.0747 ✓	−0.0915 ✓	−0.0953 ✓	−0.0910 ✓	−0.0763 ✓
1000	−0.0597 ✓	−0.0769 ✓	−0.0855 ✓	−0.0855 ✓	−0.0825 ✓

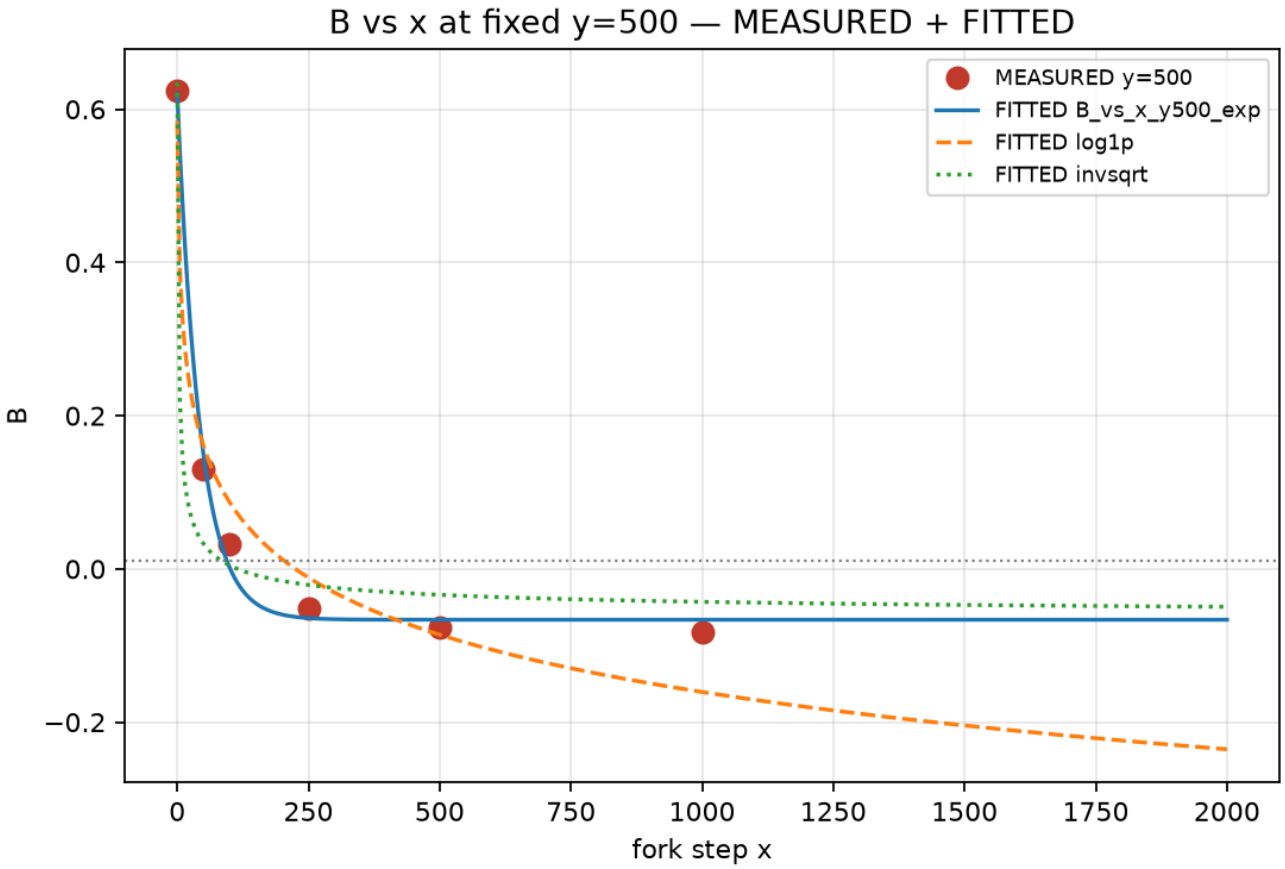


Quantitative properties (FITTED — Batch 1 grid only)

ID	Relation	Form	(R^2)	Domain
Fit A growth	(B) vs (y) @ (x=0)	(B=-0.1901+0.001628,y)	(\approx 1.000)	(x=0), (y\ge 100)
Fit A quad	(B) vs (y) @ (x=0)	(B=-0.0765+7.61\times10^{-4},y+1.30\times10^{-6},y^2)	0.991	all five (y)
Fit B	(B) vs (x) @ (y=500)	(B=-0.0663+0.6870,e^{-0.02316,x})	0.995	(y=500), six (x)
Fit C	(G) vs (B)	(G=0.0897+1.0328,B)	0.989	all 30 cells

Identity check: (G=B+(\bar L-L_{\mathrm{ens}})); MEASURED (\bar L-L_{\mathrm{ens}}) mean (\approx 0.089) → (G\approx B+0.09). Out-of-grid bands remain **PREDICTION** / Kind=`extrapolated`.


B vs y at fixed x — MEASURED + FITTED
x=0 MEASURED
x=50 MEASURED
x=100 MEASURED
x=250 MEASURED
x=500 MEASURED
x=1000 MEASURED
x=0 quadratic FITTED
B=0.01 threshold
B
post-fork steps y
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
100
200
300
400
500



(c) Interpolation curves

Every cell has (\alpha\in{0,0.25,0.5,0.75,1}) in `a3-mode-connectivity.json`.

Cell	(L(\theta(\alpha)))	(B)	(L_{\mathrm{ens}})	Shape
((0,500))	[3.7511, 3.9156, 4.3734, 3.8961, 3.7483]	0.6237	3.6454	unimodal ridge @ 0.5
((50,500))	[3.7009, 3.7345, 3.8253, 3.7265, 3.6898]	0.1299	3.5943	milder ridge
((1000,500))	[3.3812, 3.3170, 3.2986, 3.3172, 3.3810]	−0.0825	3.3017	concave (no ridge)

(d) Same-basin controls

Ablations (LR / batch / schedule): **PENDING**.
Grid implication (MEASURED, `wsd0.0` only): (B>0.01) only at small (x) + large (y); Fit B excess-barrier (\sim e^{-0.023,x}) at (y=500) is the scorecard for future schedule ablations.

Problem 2: River-Valley Dynamics

MEASURED finals: `media/p2/batch2_finals.csv` (jobs 0–6). Best raw val: linear (L^\star=2.9282028675079346). Regret (R=L-L^\star).

Mechanism / intuition. Decayed schedules shrink late bounce amplitude and let the iterate settle into the valley floor; constant `wsd0.0` keeps wall-bouncing through the full budget, so final val sits far above

decayed peers. EMA / temporary dip / horizon-free arms probe whether averaging or budget-agnostic LR can recover that gap without a tuned decay.

Job	Arm	W&B	Final val (exact)	Quantity (MEASURED)
1	linear	`r0294p93`	2.9282028675079346	(R=0) (best)
2	cos	`sktw67q8`	2.934119462966919	(R=0.005916595458984375)
3	`wsd0.2`	`soosjcxr`	2.93774151802063	(R=0.009538650512695312)
0	`wsd0.0`	`widt420l`	3.0975050926208496	(R=0.16930222511291504) (worst)
4	`wsd0.2`+dip	`rms7iyyt`	2.938987970352173	(R_{\mathrm{dip}}=0.0012464523315429688) vs job 3
5	invsqrt	`1jri34em`	3.020456075668335	(\Delta L=0.09225320816040039) vs linear
6	SGDR (T=2000)	`51b3qq4n`	3.0348048210144043	(\Delta L=0.10660195350646973) vs linear

(a) Schedule comparison

PREDICTION: decayed > constant; top-2 among `{linear, cos, wsd0.2}`; `wsd0.0` (R\in[0.02,0.08)).
MEASURED ranking: linear (<) cos (<) `wsd0.2` (≪) `wsd0.0`.

Ledger	PREDICTION	Observed	
`P2-sched-rank`	(\min L_{\mathrm{decayed}} < L_{\mathrm{wsdo.o}})	linear best; `wsd0.0` worst	**HIT**
`B2-P2-linear-R`	(R\in[0,0.02))	(0)	**HIT**
`B2-P2-cos-R`	(R\in[0,0.02)); top-2	(0.005916595458984375); #2	**HIT**
`B2-P2-wsd0.2-R`	(R\in[0,0.02)); top-2	(0.009538650512695312); #3	band **HIT**; top-2 **MISS**
`B2-P2-wsd0.0-R`	(R\in[0.02,0.08)); worst	(0.16930222511291504); worst	band **MISS**; worst **HIT**

Constant-LR regret was under-predicted by ~(2\times) the upper band edge — bounce amplitude under `wsd0.0` dominates.

(b) Eval-only EMA

PREDICTION: gap-closed fraction (f\in[0.5,0.75)) (`P2-ema-gap-frac`).
Results: PENDING — need EMA series export; do not invent.

(c) Temporary LR dip

Dip `lr_dip=(0.4,0.45,0.5,0.1)` on `wsd0.2`; control = job 3.
PREDICTION: final (R\in[-0.02,0.02)); mid-dip (R_{\mathrm{mid}}<-0.02).
MEASURED: (R_{\mathrm{dip}}=0.0012464523315429688\in[-0.02,0.02)) → `B2-P2-dip-R` **HIT**. Mid-dip: **PENDING**.

(d) Horizon-free schedules

PREDICTION: (\Delta L=L_{\mathrm{hf}}-L_{\mathrm{linear}}\in[0.01,0.08)); (\lvert L_{\mathrm{sgdr}}-L_{\mathrm{invsqrt}}\rvert\in[0,0.03)).
MEASURED: invsqrt (\Delta L=0.09225320816040039) **MISS**; SGDR (\Delta L=0.10660195350646973) **MISS**; closeness (0.014348745346069336) **HIT**. Horizon-free arms track each other but both sit ~0.1 above the tuned linear floor.

Problem 3: The Edge of Stability

Mechanism / intuition. Full-batch GD with fixed (\eta) drives (\lambda_{\max}) toward (\sim 2/\eta) (progressive sharpening → edge). Minibatch AdamW instead tracks a pre-update product (\eta\lambda^{\mathrm{pre}}) that can sit far above 2 while still training — a different "edge" diagnostic than SGD's (2/\eta). Handout thresholds (mechanism, not Observed): GD edge (\eta\lambda=2) ⇒ (\lambda=200) at (\eta=0.01); Adam (\beta_1=0.9) precond edge (38).

(a) Full-batch (Cohen-style)

Clean case (N=32), (\eta=0.01) (job 7 / W&B `0dewkykj`). η-halve/double @ 750: **PENDING**.

Quantity	PREDICTION	MEASURED	
median (\lambda_{\max}) steps 1200–1500	`[150,250)`	**245.16314737956807**	**HIT** (near (2/\eta=200))

Late train loss still decreases (final train `1.268615484237671` @ 1499; val `8.053705215454102` — small-(N) overfitting, not the edge metric). Crossing ~step 375; plateau (\eta\lambda\approx 2.55).

(b) Minibatch AdamW

Job 8 / `foew12y0` (val `3.2397050857543945` @ 2999). Primary: post-warmup median (\eta\lambda^{\mathrm{pre}}).

Quantity	PREDICTION	MEASURED	
post-warmup median (\eta\lambda^{\mathrm{pre}})	`[30,50)`	**30.987715914514**	**HIT** (near 38)

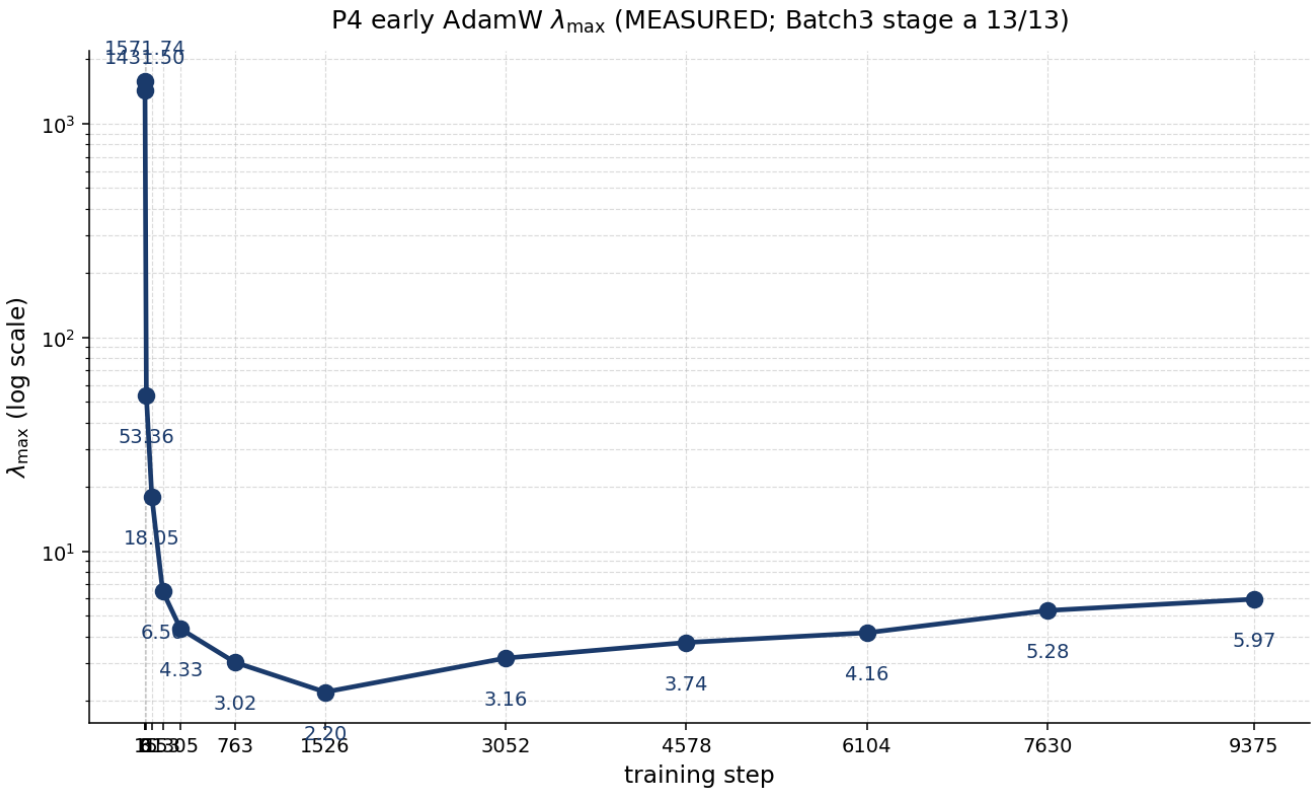
(c) Open exploration (~18 runs)

PENDING — no Batch 2/3 exploratory arms scored here.

Problem 4: Inside the Hessian

Mechanism / intuition. Over AdamW training, (\lambda_{\max}) collapses from init, dips mid-training, then rebounds into the handout edge band `[4.5,6.0]`. Family-local sharpness can migrate (embed early → head late). Negative-eigenvalue mass need not monotonically shrink. Block energy tests whether Hessian mass concentrates in architectural families vs random subspaces. Hand-rescale can move (\lambda) but typically also moves loss (invariance hard). Top-(k) SGDM subspace may carry little of the update energy.

Trajectory (MEASURED 13/13). Collapse by 15 → dip (2.1954286466134665) @1526 (earlyfit **HIT**) → rebound to (\lambda@9375=5.967310256347902) (**inside** handout `[4.5,6.0]`). Block-energy **HIT** (excess_final=0.08727119221832025; fam_above=True; layer_above=True). Plot: `media/p4/p4_lambda_max_early.png`.



MEASURED (\lambda_{\max}) card

Step	(\lambda_{\max}) (exact)
0	1431.5030691763686
3	1571.7420262270407
15	53.36426571742344
61	18.054717868295125
153	6.502395300121604
305	4.32926630998023
763	3.02398152405625
1526	2.1954286466134665
3052	3.1637529312879873
4578	3.7423705126158477
6104	4.155164815834522
7630	5.281516516770698
9375	5.967310256347902

Density (file-exact): neg_frac (0.004510114309467541\to 0.014913742516007O9\to 0.9344596369432798\to 0.06965100420936976) (steps 0→153→1526→9375). Final in `[0,0.15)` but **did not shrink** vs init → `P4-a-density-neg` **MISS**.

(a) Scorecard

Ledger ID	PREDICTION (locked)	Observed
`P4-a-family-sharpest`	Late: attention or mlp; ratio `[1.5,5)`	**MISS** — late=`head` ((5.641603135007765)); early_top2=`embedding`,`attention`; ratio=(7.38436579244244)
`P4-a-norm-sharpness`	Nondecreasing; final (\lambda_{\max}) in `[4.5,6.0]`	**HIT** — (\lambda) (2.1954286466134665\to5.967310256347902); (\mathrm{tr}(H^2)/d) (8.856225326736765\mathrm{e}{-7}\to5.364973956804868\mathrm{e}{-6})
`P4-a-density-neg`	Neg mass shrinks; final `[0,0.15)`	**MISS** — final (0.06965100420936976); shrink_0→9375=false
`P4-a-block-energy`	Family within-frac > random; excess `[0.05,0.30)`	**HIT** — excess_final=0.08727119221832025; fam_above=True; layer_above=True
`P4-lam-1526-earlyfit`	(\lambda@1526\in[1.4,4.0))	**HIT** — (2.1954286466134665)
`P4-lam-9375-earlyfit`	(\lambda@9375\in[0.3,4.0))	**MISS** — (5.967310256347902) (inside handout `[4.5,6.0]`)

MISS mechanisms (predictive — locked PREDICT kept). Late sharpest stayed `head` (not attn|mlp). Density cleanup direction failed. Earlyfit envelope undershot the rebound into the handout edge band.

(b) Move (\lambda_{\max}) without moving the loss

Baseline (MEASURED `media/p4/b-rescale-reference.json`): (\lambda_{\max} =5.967310256347902), `probe_loss`(=2.9474442899227142), `scale={}`.

Ledger ID	PREDICTION (locked)	Observed
`P4-invar-dL`	(\lvert\Delta L_{\mathrm{val}}\rvert\in[0,0.002)); (\lambda) ratio (\ge 2) or (\le 0.5)	**MISS** — `

(c) SGDM top subspace

Ledger ID	PREDICTION (locked)	Observed
`P4-top10-frac`	Mean (|P_{10}g|/|g| \in[0.6,0.85))	**MISS** — mean top_frac=0.09296940652964016
`P4-c-loss-rank`	full ≈ top < removed; gap `[0.02,0.20)`	**MISS** — L_full=3.582527816295624; L_top=5.502330541610718; L_removed=3.5821831226348877; gap=-0.000344693660736084

Problem 5 (optional): Own prediction problem

Phenomenon: after early fork under `wsd0.0` already yields large (B), can a short matched cooldown collapse (B\le 0.01) (same valley walls) vs constant-LR control (true modes)?

Baseline (MEASURED P1): ((0,500)) (B=0.6237); ((50,500)) (B=0.1299); ((1000,500)) (B=-0.0825).

ID	PREDICTION	Observed
P5-B-cool-x0	(B_C\le 0.01)	**MISS** — B_C=0.7697049379348755
P5-dB-cool	(B_R-B_C\in[0.5,0.8))	**MISS** — dB=-0.14600493793487546 (B_R=0.6237 – B_C=0.7697049379348755)
P5-B-const-ctrl	(B_K\ge 0.5)	**HIT** — B_K=0.7320129871368408
P5-rank-arms	(B_C < B_R) and (B_C < B_K)	**MISS** — B_C=0.7697049379348755, B_R=0.6237, B_K=0.7320129871368408
P5-G-cool-fitC	if (B_C>0.01): (\lvert G-(B+0.09)\rvert<0.05)	**HIT** — G_C=0.8391239460323332; |G−(B+0.09)| =0.020580991902542256
P5-emergent-basin	Same-valley if (B_C\le 0.01) and (B_K>0.01)	**MISS** — JSON emergent='true-modes'
P5-L-cool-dir	(L_C<L_R); (\Delta L\in[0.02,0.15))	**INSUFFICIENT** — dL null / (L_R,L_C) missing

Takeaway. Short matched cooldown failed to create connectivity on the worst P1 cell; constant-LR control stayed high-(B); Fit C still held; forced choice is true disconnected modes, not opposite walls of one valley. Code: `experiments/a3_optimization/p5_cooldown_barrier.py`. Media: `media/p5/`.

Interpretation summary

P1 (MEASURED + FITTED). Weight averaging works on 26/30 cells. Failures = small-(x), large-(y) under constant LR; worst (B=0.6237311363220215) at ((0,500)). Ensemble still wins there ((G) up to 0.7281). Fits A–C summarize growth in (y), exponential collapse in shared (x), and (G\approx B+0.09).

P2 (MEASURED 0–6). Decayed schedules beat constant (**HIT** rank); `wsd0.0` regret (0.16930222511291504) under-predicted (**MISS** band). Dip final ≈0 (**HIT**). Horizon-free (\Delta L\sim 0.09)–(0.11) vs linear (**MISS** bands) but track each other (**HIT** closeness). EMA / mid-dip still **PENDING**.

P3 (MEASURED jobs 7–8). SGD late (\lambda_{\max}=245.16314737956807) → **HIT** `[150,250)`. AdamW post-warmup (\eta\lambda^{\mathrm{pre}}=30.987715914514) → **HIT** `[30,50)`. η-change / ~18-run exploration **PENDING**.

P4 (MEASURED 13/13 + b/c). Rebound to (\lambda@9375=5.967310256347902) (**inside** handout `[4.5,6.0]`). Predictive MISSes kept locked: late sharpest=`head`; dens-neg no shrink; earlyfit@9375 past `[0.3,4.0)`. Norm **HIT**; block-energy **HIT** (excess_final=0.08727119221832025). `P4-invar-dL` **MISS** (|ΔL| =2.73853462934494); `P4-top10-frac` **MISS** (mean=0.09296940652964016); `P4-c-loss-rank` **MISS** (gap=-0.000344693660736084).

P5. Cooldown did not collapse barrier (`P5-B-cool-x0` / `P5-dB-cool` / `P5-rank-arms` / `P5-emergent-basin` **MISS**); constant-LR control stayed high-(B) (**HIT**); Fit C held (**HIT**); `P5-L-cool-dir` **INSUFFICIENT**.

Compact quantitative card (all MEASURED / FITTED / scored)

P2 scored counterfactuals

ID	PREDICTION	Observed (exact)	Score
`B2-P2-dip-R`	([-0.02,0.02))	(0.0012464523315429688)	**HIT**
`B2-P2-invsqrt-dL`	([0.01,0.08))	(0.09225320816040039)	**MISS**
`B2-P2-sgdr-dL`	([0.01,0.08))	(0.10660195350646973)	**MISS**
|SGDR−inv|	([0,0.03))	(0.014348745346069336)	**HIT**

PENDING (do not invent)

Topic	Status
Mid-dip (R_{\mathrm{mid}})	**PENDING**
EMA gap (`P2-ema-gap-frac`)	**PENDING**
P3 η-change / P3(c)	**PENDING**
`P5-L-cool-dir`	**INSUFFICIENT** (dL=null)

Appendix highlight — P1 full cell table (MEASURED)

x	y	L(θ₁)	L(θ₂)	L_ens	L(θ_avg)	B
0	25	6.4380	6.4653	6.4077	6.4228	−0.0288 ✓
0	50	5.6329	5.6864	5.5798	5.6180	−0.0416 ✓
0	100	5.0282	5.0396	4.9238	5.0064	−0.0275 ✓
0	250	4.3202	4.3092	4.2046	4.5319	0.2173
0	500	3.7511	3.7483	3.6454	4.3734	0.6237
50	25	5.2708	5.2877	5.1962	5.1784	−0.1009 ✓
50	50	5.0453	5.0449	4.9441	4.9398	−0.1053 ✓
50	100	4.6943	4.6940	4.5872	4.6031	−0.0910 ✓
50	250	4.1278	4.1245	4.0231	4.1316	0.0055 ✓
50	500	3.7009	3.6898	3.5943	3.8253	0.1299
100	25	4.8268	4.8219	4.7374	4.7311	−0.0932 ✓
100	50	4.6870	4.6719	4.5790	4.5776	−0.1019 ✓
100	100	4.4887	4.4439	4.3591	4.3691	−0.0972 ✓
100	250	3.9961	3.9747	3.8808	3.9342	−0.0512 ✓
100	500	3.6670	3.6582	3.5634	3.6945	0.0319
250	25	4.2400	4.2449	4.1581	4.1461	−0.0963 ✓
250	50	4.1383	4.1451	4.0505	4.0311	−0.1106 ✓
250	100	3.9972	3.9767	3.8869	3.8709	−0.1161 ✓
250	250	3.7429	3.7357	3.6447	3.6615	−0.0779 ✓
250	500	3.5740	3.5712	3.4844	3.5204	−0.0522 ✓
500	25	3.7266	3.7154	3.6522	3.6463	−0.0747 ✓
500	50	3.7105	3.6971	3.6226	3.6123	−0.0915 ✓
500	100	3.6654	3.6611	3.5798	3.5680	−0.0953 ✓
500	250	3.5767	3.5748	3.4918	3.4847	−0.0910 ✓
500	500	3.4862	3.4857	3.4037	3.4097	−0.0763 ✓
1000	25	3.4786	3.4796	3.4221	3.4194	−0.0597 ✓
1000	50	3.4730	3.4757	3.4051	3.3974	−0.0769 ✓
1000	100	3.4619	3.4611	3.3867	3.3760	−0.0855 ✓
1000	250	3.4241	3.4225	3.3449	3.3378	−0.0855 ✓
1000	500	3.3812	3.3810	3.3017	3.2986	−0.0825 ✓

Full configs / W&B IDs / family-(\lambda) tables remain in source MD (`a3-experimental-appendix.md`) — not re-dumped here.